

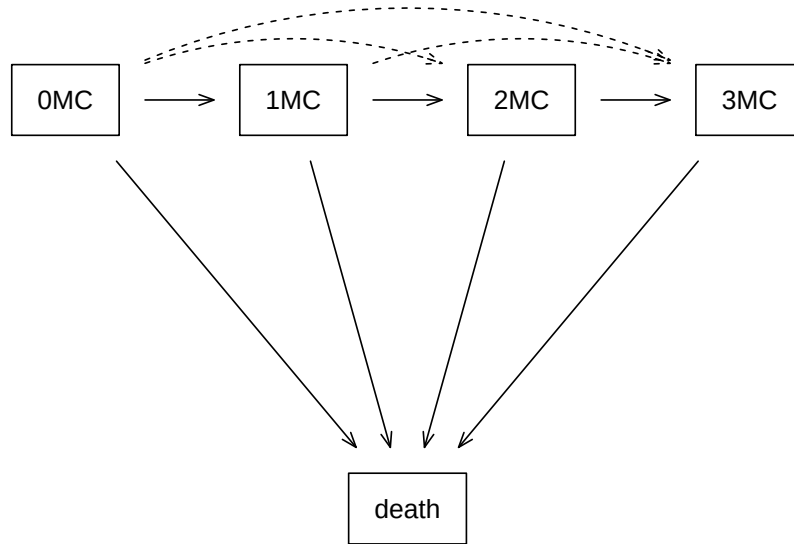
# Some open questions

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## 1 Fatty liver disease

In the analysis of the fatty liver disease data (NAFLD) had the following biological model.



For the analysis I had used a multi-state hazard (MSH) model that was event based. That is, it used coxph, and the underlying data were the times of transitions plus baseline covariates (sex, NAFLD, BMI, ...). But the analysis raises some questions.

A primary one is that the study is based on medical history data, where visits are a random process (random in the sense that we don't know when or why a particular subject will return). Using the availability of an HDL value as a surrogate for a visit that "would have recorded" the 3 metabolic comorbidities if they were present, the code below shows the distribution of inter-visit times.

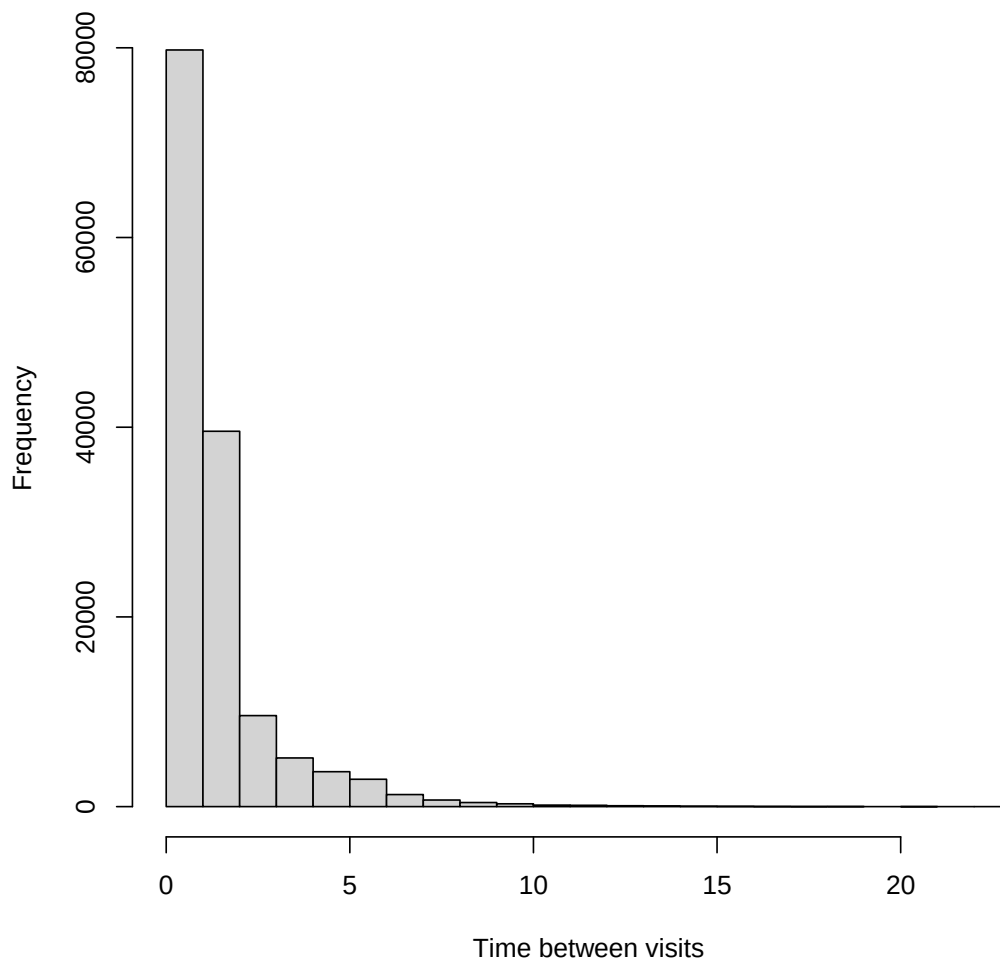
```

hdldat <- unique(subset(nafld2, test=="hd1"))
temp <- with(hdldat, tapply(days, id, diff))
delta <- unlist(temp)
quantile(delta)

##    0%   25%   50%   75%  100%
##     1   142   339   502  8113

hist(delta[delta>7]/365.25, nclass=30, main=NULL, xlab="Time between visits")

```



Question 1: Clearly the true date of an incident MC is on or before the detected date, perhaps by several years. What is the impact of this on estimated coefficients? What is the impact on the estimated absolute risk?

Question 2: Some subjects who were recruited as controls were later diagnosed with NAFLD. What is the impact of different ways of dealing with this?

- Ignore the change, and only use the initial cases and control
- Treat NAFLD as a time-dependent covariate
- Treat NAFLD as a state; we would now have  $2*4 + \text{death} = 9$  states

Question 3 (harder): It is known from NHANES and other studies that the population prevalence of fatty liver is 20–30%. The NAFLD study identified 3851 cases in a 7 year window, in an Olmsted County population of roughly 100 thousand = 3.8%. This means that a portion of our 'controls' have undetected NAFLD. What is the impact of this?

## 2 Mayo Clinic Study of Aging

In the 2022 paper we looked at the impact of amyloid on the incidence of dementia (hazard ratios), current status ( $p(t)$ ), lifetime risk of ever having dementia, and expected total years in the dementia state. The data set used study visits, off study record review, and last follow-up status information. As a reminder, the study protocol specifies a study visit every 15 months. It is not common but also not rare for a patient to skip a study visit sometime in their tenure, what is common is for someone to leave the study without announcing their intention to do so. If a subject has not been seen in 36 months, their identifier is put on the review list. Study nurses then retrieve the medical history and review it for dementia, which can be diagnosed via a standard MD visit; but MCI can not be distinguished. A status of death or last alive comes from automated systems which are based on much broader data sources.

Question 4: The fact that there were 395 on study dementias and 430 off study dementias is prima facie evidence of differential dropout (and it is true). What is the impact of ignoring the off study data? (Many large studies in this field do not have access to such data.) Hypothesis: it has a bigger effect on absolute risk than on the hazard ratio.

Question 5: Only about 40% of the subjects participate in imaging (.37 in the public data set); the biggest reason for non-participation appears to be time – imaging requires a second appointment which normally takes 2 full days. In the 2022 paper we categorized amyloid as low/moderate/high/missing and used all subjects, under the assumption that we would have better precision for the male, e4, and cmc effects.

- Is that true?
- What is the impact on the absolute rates, of using only the amyloid subjects?

Question 6: In a reprise of question 1, what is the impact of interval censored visit times? On study visits are supposed to be every 15 months, a schedule which patient adhere to imperfectly. History review occurs at 3 year intervals.

Note: There is significant non-proportional hazards for the female APOEε4 effect, which we ignored. What is a better estimate of that effect, and what is its impact on the absolute rates? A preliminary look implied that the impact of this non-PH is quite disparate.

### 3 Crohn's disease

Many of the above questions apply to the data set on Crohn's disease, which had a rather ad hoc analysis for the study paper (I was new to MSH models). What does a proper analysis look like, and what can we learn?

### 4 PBC

The sequential PBC data set was used as a platform to explore the idea of replacing a time dependent covariate with a set of states in a MSH model. The first look showed that the idea might have merit, but more depth is needed. Such a look does not necessarily need to use the PBC data set. Cortese and Andersen (2009) looked at time-dependent categoricals.

Question 7: We divided bilirubin into 3 groups of normal ( $\leq 1$ ), 1–4 and 4+. What is the proper number of groups, and how could one pick cutpoints?

Question 8: What if there are multiple TD covariates. Consider bilirubin and albumin in the PBC data, for instance. One could divide each into 3 groups to get  $3*3 + \text{death} = 10$  states. How many groups, and many covariates can we handle?

Question 9: The book chapter 14 implies that a MSH model (coxph) can handle more groups than an AJ, but imposing proportional hazard functions. True?

Question 10: An alternate way to deal with time dependent covariates is the joint modeling framework. How does this approach compare?

### 5 Interval censored MSH models

Interval censored hazard models seem to have promise as a way to address many of the questions above. A contrast of event based MSH (coxph) or interval censored MSH (mshmm) can be explored for any of them.